

Offline line-art vectorization v0.99.4

Turn a scanned figure/schematic into crisp SVG — razor-sharp at ANY deep-zoom, clean to reprint or recolour. No CDN, no external binary.

THE PIPELINE (engine/vectorize.py)



WHY IT PAIRS WITH DEEP-ZOOM

Infinitely crisp
A raster blurs when you zoom past its DPI; an SVG is resolution-free — the browser renders the Vectorize output

Offline, uses cv2 already present
OpenCV ships with the OCR stack, so no new dependency. Cached per page; if cv2 is missing the route returns a clean

Reprint / recolour clean
Vector line-art prints crisp and can be recoloured (it's a filled path) — useful for take-to-the-bay packets.

Keeps the fine detail
Tuned so hatching, thin leader lines and callout text survive (no median-blur, low min-area).

VERIFIED

- Core logic run in-sandbox on synthetic line-art: 18 contours, faithful reproduction incl. hatching + text.
- Proof docs/vectorize_proof.png (raster -> vector, side by side). OpenCV 4.13 + numpy + Pillow.
- Route (/vectorize) + button pending the host suite (VERIFY-099.bat) — the mount truncated the grown files.
- Additive & rollbackable (R1): one module + one route + one button; veccache sidecar only. Closes thread #4 fully.